

👉 XAI Lecture 13

Explainable Medical Coding & Attention is not Explanation

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Explainable Prediction of Medical Codes from Clinical Text

Authors: J. Mullenbach, S. Wiegrefe, J. Duke, J. Sun, J. Eisenstein

Presented by: Leo Benac, Chelse Swoopes, Kelly Zhang

[@mullenbach2018explainable]

🍷 Introduction ---

- **Clinical notes**: free text narratives generated by clinicians during patient encounters
- **ICD codes**: metadata codes accompanying clinical notes - Standardized way of indicating diagnoses and procedures - Multiple use cases (billing, predictive modelling)





Motivation ---

- Limitation of ICD code annotation
 - ▶ Labor intensive
 - ▶ Coders read 720–1,440 pages of clinical documentation per day
 - ▶ Prone to error
 - ▶ Lack of connection bridging text and code
- Challenges of automatic coding
 - ▶ Features of clinical text: irrelevant information, misspellings, non-standard abbreviations, large medical vocabulary
 - ▶ Label space

Summary of Contributions ---

- **CAML**: Convolutional Attention for Multi-Label classification
 - ▶ Attentional convolutional network: CNN + per-label attention mechanism
 - ▶ Important information may be in short snippets anywhere in the document
 - ▶ Decision support in clinical domain
- **Accuracy Evaluation**
 - ▶ Used open-access MIMIC datasets
 - ▶ Outperformed previously proposed approaches
- **Interpretability Evaluation**
 - ▶ Physician rated the informativeness of automatically generated explanations

🍷 Related Work: Automatic ICD Coding ---

- Structured text v. unstructured text - ****Datasets****
- Subset of full ICD label space (Wang et al., 2016) - Subset of medical scenarios (Zhang et al., 2017) - Death certificates in English + French (Névéol et al., 2017) - Private datasets (Subotin and Davis, 2016) - ****Approaches**** - Recurrent networks with HA-GRU (Baumel et al., 2018) - Character-aware LSTMs (Shi et al., 2017) - Memory networks for top-50/100 codes (Prakash et al., 2017) - Grounded RNN for individual labels (Vani et al., 2017)



CAML yields stronger results across all experimental conditions.



Related Work: Explainable Text Classification ---

- Model “rationales” through a latent variable (Lei et al., 2016)
 - ▶ Tag each word as relevant to the document label
- Compute salience of individual words (Li et al., 2016)
- Use attention to highlight salient features of the text (Rush et al., 2015; Rocktäschel et al., 2016)
 - ▶ Lack interoperability evaluations of features selected by the attention mechanism

🍷 Related Work: Attentional Convolution for NLP ---

- CNNs applied to sentiment classification (Kim, 2014) and language modeling
- Convolution and attention: Allamanis et al., 2016; Yin et al., 2016; dos Santos et al., 2016; Yin and Schütze, 2017
- Most relevant work (Yang et al., 2016 and Allamanis et al., 2016):
 - ▶ Use context vectors to compute attention over specific locations in the text
- **Differentiation**
 - ▶ Goal: select locations most important for predicting specific labels
 - ▶ Compute separate attention weights for each label in the label space

🍷 ICD Code Prediction Problem ---



- **Multi-Label Classification Problem****
- "Match" snippets of clinical text to a label -
Use code descriptions to help make text snippets relevant to predicted codes (esp. for labels with little data)

🍷 Typical Convolutional N-Gram Model ---



- Word embeddings

🍷 Typical Convolutional N-Gram Model ---



- Max pooling → Was the “N-gram” detected at all in this piece of text?

🍷 Convolutional Attention Model ---



- $\mathbf{X}$: text word embeddings, dimension d_e , length N
- $\mathbf{H}$: N-gram embeddings, dimension d_c , length N
- Convolution maps $\mathbf{X} \rightarrow \mathbf{H}$

🍷 Convolutional Attention Model ---



$$\alpha_{\ell} = \text{SoftMax}(\mathbf{H}^{\top} \mathbf{u}_{\ell})$$

Weight representing "match" between n-grams
in document ($\mathbf{H}$) and label embedding ($\mathbf{u}_{\ell}$)

🍷 Convolutional Attention Model ---



$$\alpha_{\ell} = \text{SoftMax}(\mathbf{H}^{\top} \mathbf{u}_{\ell})$$

$$\mathbf{v}_{\ell} = \sum_{n=1}^N \alpha_{\ell,n} \mathbf{h}_n$$

N-grams weighted by α form document vector
 $\mathbf{v}_{\ell}$

🍷 Convolutional Attention Model ---



$$\alpha_\ell = \text{SoftMax}(\mathbf{H}^\top \mathbf{u}_\ell)$$

$$\mathbf{v}_\ell = \sum_{n=1}^N \alpha_{\ell,n} \mathbf{h}_n$$

$$\hat{y}_\ell = \sigma(\beta_\ell^\top \mathbf{v}_\ell + b_\ell)$$

Training Loss Function ---

$$L_{\text{BCE}}(\mathbf{X}, y) = - \sum_{\ell=1}^{\mathcal{L}} [y_{\ell} \log(\hat{y}_{\ell}) + (1 - y_{\ell}) \log(1 - \hat{y}_{\ell})]$$

Binary Cross-Entropy Loss

Binary classification for each possible label

$$L(\mathbf{X}, y) = L_{\text{BCE}} + \lambda \frac{1}{n_y} \sum_{\ell: y_{\ell}=1} \|\mathbf{z}_{\ell} - \beta_{\ell}\|_2$$

Regularization

Make label representations close to code descriptions

🍷 Training Loss Function ---

$$L(\mathbf{X}, y) = L_{\text{BCE}} + \lambda \frac{1}{n_y} \sum_{\ell: y_\ell=1} \|\mathbf{z}_\ell - \beta_\ell\|_2$$

- n_y : number of true labels for data example
- $\mathbf{z}_\ell$: representation of label based on code description
- β_ℓ : representation of label learned by model

Dataset ---

- **MIMIC-3**

- Input: sequence of words describing a patient's ICU stay (max 2500 words)
- Output: ICD-9 codes for diagnoses and procedures
- 9,000 unique ICD-9 codes: 7,000 diagnosis + 2,000 treatment codes
- Preprocessing: 100-dimension embeddings

	MIMIC-III full
# training documents	47,724
Vocabulary size	51,917
Mean # tokens per document	1,485
Mean # labels per document	15.9
Total # labels	8,922

Baselines ---

- Single Layer CNN
- Logistic Regression
- Bidirectional Gated Recurrent Unit

Evaluation Metrics ---

- Micro-averaged and Macro-averaged F1 and AUC
- Precision@n — most common n labels in ground truth data

$$\text{Micro-R} = \frac{\sum_{\ell=1}^{|\mathcal{L}|} \text{TP}_{\ell}}{\sum_{\ell=1}^{|\mathcal{L}|} \text{TP}_{\ell} + \text{FN}_{\ell}}$$

$$\text{Macro-R} = \frac{1}{|\mathcal{L}|} \sum_{\ell=1}^{|\mathcal{L}|} \frac{\text{TP}_{\ell}}{\text{TP}_{\ell} + \text{FN}_{\ell}}$$

🍷 Results ---

Model	AUC Macro	AUC Micro	F1 Macro	F1 Micro	P@8	P@15
Scheurwegs et al. (2017)	–	–	–	–	–	–
Logistic Regression	0.561	0.937	0.011	0.272	0.542	0.411
CNN	0.806	0.969	0.042	0.419	0.581	0.443
Bi-GRU	0.822	0.971	0.038	0.417	0.585	0.445
CAML	0.895	0.986	0.088	0.539	0.709	0.561
DR-CAML	0.897	0.985	0.086	0.529	0.690	0.548

- CAML gives strongest results on all metrics; DR-CAML is the regularized version ($\lambda = 0.01$)
- Bi-GRU comparable to vanilla CNN; Logistic Regression worse than all neural architectures

🍷 Interpretability Evaluation ---

- Sample of 100 predicted codes
- Most important k -gram ($k = 4$) + five words on each side for context
- Asked the Physician how informative it is

🍷 Interpretability: Qualitative Examples ---



🍷 How to Pick the Most Influential 4-gram? ---

- **CAML**: looks at SoftMax output α_ℓ to get the best 4-gram for code ℓ
- **Logistic Regression**: sum of weight matrix coefficients for ℓ over words in k -gram; top-scoring k -gram returned
- **Code descriptions**: cosine similarity between each stemmed k -gram and stemmed ICD-9 code description
- **Max-Pooling CNN**:

$$a_i = \arg \max_{j \in \{1, \dots, m-k+1\}} (H_{ij}), \quad \alpha_{i\ell} = \sum_{j: a_j=i} \beta_{\ell,j}$$

Select the most important k -gram as $\arg \max_i \alpha_{i\ell}$

Qualitative Evaluation Results ---

Method	Informative	Highly informative
CAML	46	22
Code Descriptions	48	20
Logistic Regression	41	18
CNN	36	13

- CAML selects the greatest number of “highly informative” explanations
- CAML selects more “informative” explanations than both CNN and logistic regression
- Cosine Similarity also performs well; examples in Table 1 demonstrate strengths of CAML

Conclusion ---

Concluding Thoughts

- Yields strong improvements over previous metrics across several formulations of ICD-9 code prediction
- Provides satisfactory explanations for its predictions per stakeholder evaluation

Future Work

- **Extensible** to other multi-label document tagging tasks
- **Linguistic perspective**: integrate document structure of discharge summaries; better handle out-of-vocabulary tokens
- **Application perspective**: leverage hierarchy of ICD codes; predict codes for future visits

Discussion (Paper 1) ---

- ① What are potential limitations of CAML?
- ② The explanations from the attention mechanism are extracted text snippets. Does this provide enough information? What else could be provided?
- ③ Strengths and weaknesses of their evaluation and results (accuracy and interpretability)
- ④ What additional applications could CAML be useful for?

Attention is *not* Explanation

Authors: Sarthak Jain, Byron C. Wallace

Presenters: Nikhil Nayak, Sree Harsha Tanneru, Hongjin Lin

2023/03/06

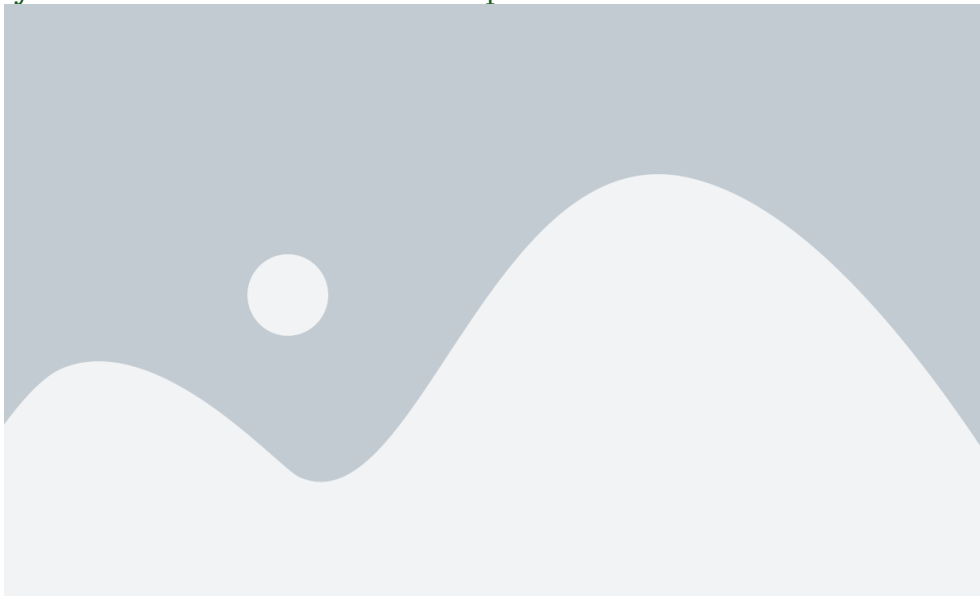
[@jain2019attention]

🍷 Attention Mechanism: a "Recent" Breakthrough in NLP ---

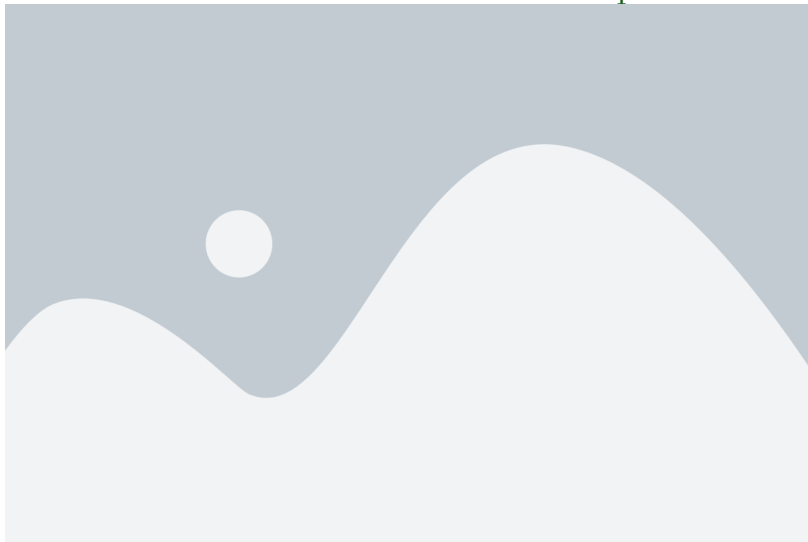
- **2015**: birth of attention mechanism in NLP
 - ▶ “Neural machine translation by jointly learning to align and translate” (Bahdanau et al., 2015)
- **2017**: birth of transformers based on self-attention mechanism
 - ▶ “Attention is all you need” (Vaswani et al., 2017)



🍷 Many Use Attention as an Explanation Mechanism ---



🍷 But Does Attention Provide Faithful Explanations? ---



Despite very different attention weights, both yield effectively the same prediction (0.01).

What is the degree to which attention weights provide meaningful “explanations” for predictions?

- 1 **Consistency:** To what extent do induced attention weights **correlate with measures of feature importance** — specifically, those from gradients and leave-one-out (LOO) methods?
- 2 **Counterfactual attention weights:** Would **alternative attention weights** (and hence distinct heatmaps/“explanations”) necessarily yield different predictions?

🍷 Contributions and Main Findings ---

- ① Standard attention modules **do not provide meaningful explanations** and should not be treated as though they do.
- ② Learned attention weights are frequently **uncorrelated with gradient-based** measures of feature importance.
- ③ One can identify very **different attention distributions** that nonetheless yield equivalent predictions.

🍷 Experimental Setup ---



🍷 Correlation Between Attention and Feature Importance ---

To what extent do induced attention weights correlate with measures of feature importance?

Specifically two methods are considered:

- ① Feature gradient methods
- ② Leave-One-Out (LOO) method

🍷 Gradient-Based Feature Importance ---

- Compute the gradient of the output with respect to each input feature
- This tells us how much changing each feature would affect the output
- Visualize by highlighting input features with highest absolute gradient values

🍷 Feature Erasure / L00 ---

- Remove one input feature at a time and observe how the output changes
- This allows us to measure the impact of each feature on the output
- Visualize by highlighting the removed feature and showing the change in output

Distribution Change Measure ---

Total Variation Distance (TVD), Kendall's τ coefficient

$$\text{TVD}(\hat{y}_1, \hat{y}_2) = \frac{1}{2} \sum_{i=1}^{|\mathcal{Y}|} |\hat{y}_{1i} - \hat{y}_{2i}|$$

$$\tau = \frac{(\text{concordant pairs}) - (\text{discordant pairs})}{(\text{number of pairs})} = 1 - \frac{2(\text{discordant pairs})}{\binom{n}{2}}$$

🍷 Algorithm for Feature Importance Computations ---

$$\mathbf{h} \leftarrow \text{Enc}(\mathbf{x}), \quad \hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$$

$$\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \alpha)$$

$$g_t \leftarrow \left| \sum_{w=1}^{|V|} \mathbf{1}[\mathbf{x}_{tw} = 1] \frac{\partial y}{\partial \mathbf{x}_{tw}} \right|, \quad \forall t \in [1, T]$$

$$\tau_g \leftarrow \text{Kendall-}\tau(\alpha, g)$$

$$\Delta \hat{y}_t \leftarrow \text{TVD}(\hat{y}(\mathbf{x}_{-t}), \hat{y}(\mathbf{x})), \quad \forall t \in [1, T]$$

$$\tau_{\text{loo}} \leftarrow \text{Kendall-}\tau(\alpha, \Delta \hat{y})$$

🍷 Feature Gradients Results ---

Histogram of Kendall correlation between attention and gradients.



🍷 Correlation Comparison: Attention, Gradients, L00 ---



🍲 Possible Limitations in Experiment Results ---

- ➊ Potential issues with Kendall τ correlation — many irrelevant features may add noise to the correlation measures
- ➋ Alternative measures of feature importance (gradients and LOO) may not be considered ideal or the 'ground truth' for comparison purposes

🍷 Counterfactual Attention Weights ---

Would the prediction be different if the model attended to different input features?

1. **Consistency**: correlation with other measures of feature importance
2. **Counterfactual attention distributions** should yield corresponding changes in predictions



🍷 Counterfactual Attention: Two Experiments ---

- ① **Attention Permutation:** scramble the original attention weights
- ② **Adversarial Attention:** generate an adversarial distribution that is maximally distinct from the observed attention weights, whilst simultaneously close to the observed prediction

Attention Permutation: Algorithm ---

$\mathbf{h} \leftarrow \text{Enc}(\mathbf{x}), \hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$

$\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \hat{\alpha})$

for $p \leftarrow 1$ to 100 **do**

$\alpha^p \leftarrow \text{Permute}(\hat{\alpha})$

$\hat{y}^p \leftarrow \text{Dec}(\mathbf{h}, \alpha^p)$

$\Delta \hat{y}^p \leftarrow \text{TVD}[\hat{y}^p, \hat{y}]$

end for

$\Delta \hat{y}^{med} \leftarrow \text{Median}_p(\Delta \hat{y}^p)$

- $\mathbf{h}$ (the hidden state) is kept fixed - Only the attention weights are permuted - Measures how much the output changes when attention is scrambled
- ▷ $\mathbf{h}$ not changed

🍷 Attention Permutation: Results ---



Adversarial Attention ---

- Explicitly seek attention weights that differ as much as possible from observed weights, yet leaving prediction almost unchanged
- **Why?** Alternative attention distributions for the same output may be viewed as equally plausible explanations:
 - ▶ Cannot conclude the model made a specific prediction because it attended in a certain way
 - ▶ Lack of specificity makes constructing counterfactual explanations hard
 - ▶ Makes model less trustworthy



Adversarial Attention: Optimisation ---

$$JSD(\alpha_1, \alpha_2) = \frac{1}{2} \left(KL \left[\alpha_1 \mid \frac{\alpha_1 + \alpha_2}{2} \right] + KL \left[\alpha_2 \mid \frac{\alpha_1 + \alpha_2}{2} \right] \right)$$

Seek $\alpha^{(1)}, \dots, \alpha^{(k)}$ that maximize f subject to small output change:

$$\underset{\alpha^{(1)}, \dots, \alpha^{(k)}}{\text{maximize } f \left(\{\alpha^{(i)}\}_{i=1}^k \right)} \quad \text{s.t.} \quad \forall i \text{ TVD}[\hat{y}(\mathbf{x}, \alpha^{(i)}), \hat{y}(\mathbf{x}, \hat{\alpha})] \leq \epsilon$$

$$f \left(\{\alpha^{(i)}\}_{i=1}^k \right) = \sum_{i=1}^k JSD[\alpha^{(i)}, \hat{\alpha}] + \frac{1}{k(k-1)} \sum_{i < j} JSD[\alpha^{(i)}, \alpha^{(j)}]$$

🍷 Adversarial Attention: Algorithm ---

$\mathbf{h} \leftarrow \text{Enc}(\mathbf{x}), \hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$

$\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \hat{\alpha})$

$\alpha^{(1)}, \dots, \alpha^{(k)} \leftarrow \text{Optimize Eq 1}$

for $i \leftarrow 1$ to k **do**

$\hat{y}^{(i)} \leftarrow \text{Dec}(\mathbf{h}, \alpha^{(i)})$

▷ $\mathbf{h}$ is not changed

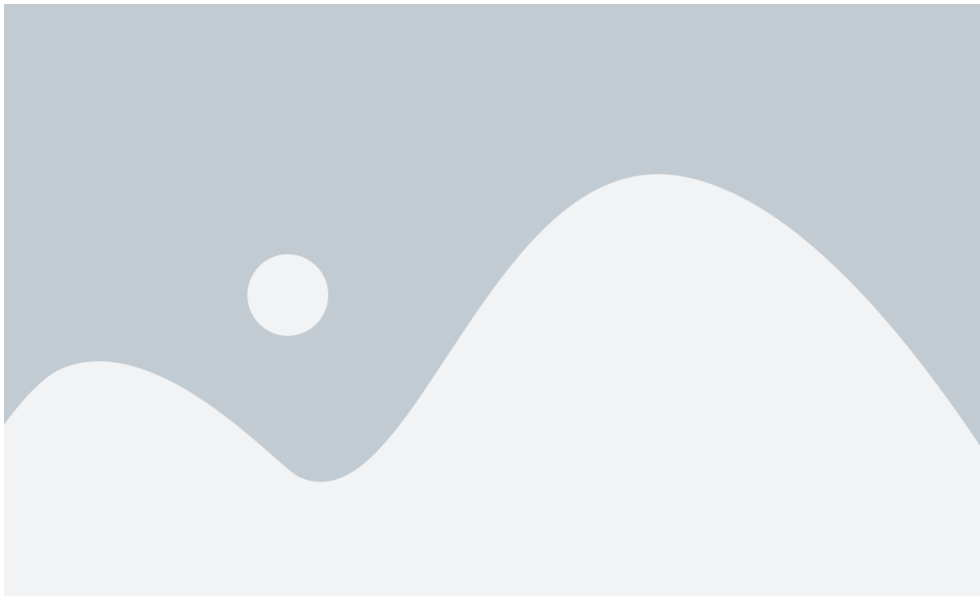
$\Delta \hat{y}^{(i)} \leftarrow \text{TVD}[\hat{y}, \hat{y}^{(i)}]$

$\Delta \alpha^{(i)} \leftarrow \text{JSD}[\hat{\alpha}, \alpha^{(i)}]$

end for

$\epsilon\text{-max JSD} \leftarrow \max_i \mathbf{1}[\Delta \hat{y}^{(i)} \leq \epsilon] \Delta \alpha^{(i)}$

🍷 Adversarial Attention: Results ---



🍷 Adversarial Attention: Results ---





Counterfactual Attention: Limitations ---

- Adversarial weights may themselves be unlikely under model parameters
- There are instances where a single explanation out of multiple plausible explanations is sufficient

Overall Limitations ---

- Alternate attention mechanisms are not explored
- Limited evaluations to classification, QA, and NPI tasks; seq2seq tasks are not considered

Critique and Discussion ---

- Wiegrefe & Pinter (2019), *Attention is not not Explanation*, EMNLP-IJCNLP:
 - ▶ Attention distribution is not a primitive
 - ▶ Is attention necessary for prediction?
 - ▶ Is the variance in attention distributions unusual? i.e., How adversarial are the adversaries?
- Do you agree with the authors' conditions of a "faithful explanation"?
- Do you think attention constitutes a "sufficient" explanation?
- What other conditions should we consider as a good explanation?
- How can experts be brought in to evaluate conditions of good explanations?

Thank You!

References --- |